

- 4 (a) The re-drawn Fig. 4.1 is as follows i.e. R_1 , R_3 and (R_2 , R_4 in series) are in parallel across X and Y.

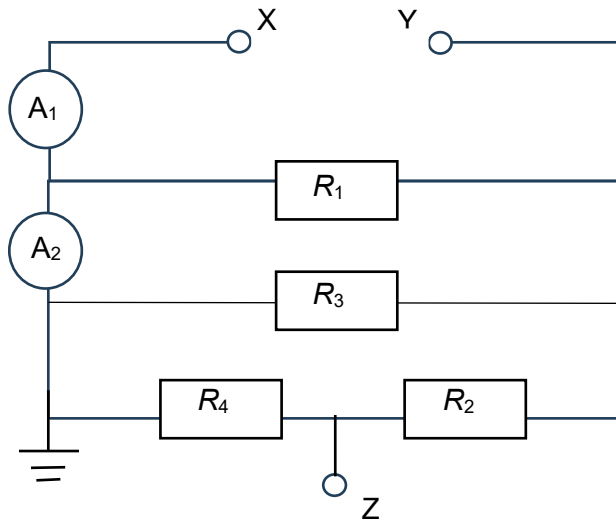


Fig. 4

The effective resistance is $\left(\frac{1}{R} + \frac{1}{R} + \frac{1}{2R}\right)^{-1} = 0.4R$

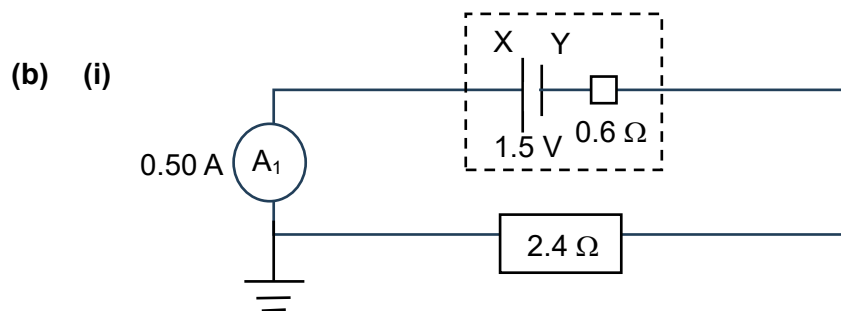
M1

$$0.4R = 2.4$$

$$R = 6.0 \, \Omega$$

A0

MC: *There are different ways to view the re-drawn Fig. 4 to solve the (b)(i) to (iii). Below is the working that should be clear enough for you to understand.*



$$\text{Total resistance} = 2.4 + 0.6 = 3.0 \, \Omega$$

M1

$$\text{Current in } A_1 = \frac{E}{R} = \frac{1.5}{3.0} = 0.50 \, \text{A}$$

A1

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- (ii) Terminal potential difference between X and Y

$$= \frac{2.4}{3.0} \times 1.5 = 1.2 \text{ V}$$

M1

$$\text{Current through resistor } R_1 = \frac{V}{R} = \frac{1.2}{6.0} = 0.20 \text{ A}$$

B1

$$\text{Thus, current in } A_2 = 0.50 - 0.20 = 0.30 \text{ A}$$

A1

- (iii) Potential difference across $R_4 = \frac{6.0}{6.0 + 6.0} \times 1.2 = 0.60 \text{ V}$

M1

$$\text{Potential at Z} = 0 - 0.60 = -0.60 \text{ V}$$

A1

- (iv) As reading is zero, the potential difference across S is 0.6 V.

C1

$$\text{The potential difference across S} = \frac{S}{S + 0.60} \times 1.5 = 0.60 \text{ V}$$

$$\text{Hence } S = 0.4 \Omega$$

A1

5(a)(i)	The magnetic force acting on the charged particle provides the	
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